SCI-VAL

Sample and Detector Validity, Flags, and Map Eligibility

Producer Facts and Use-Specific Eligibility:
Not Final Map Validity

Engineering Conformance Specification

Scientific contract version	v0.1
Document revision	r0.3
Date	2026-08-21
Formal authority	Six canonical modules in `src/common/`
Architecture	SCI-VAL Core plus owner-bound Profile Registry
Status	Owner-approved revision architecture; scientific authority not frozen; implementation conformity and validation not assessed.

Authority boundary. VAL remains the evaluator and registry mechanism, not the policy author. A registry record binds a policy to its actual owner and source without transferring authority. This document selects no PTC, MAP, or production policy.

How to use this specification

The six imported modules are the complete formal authority for this revision. The separate science-team rationale explains the same architecture without reproducing tuples, truth tables, or numbered clauses. Every numbered requirement is an observable obligation and every prediction is a potential falsifier. No clause reports a test result.

An engineering review must treat producer facts, registry records, owner profiles, evaluated decisions, consumer actions, and lifecycle generations as separate identities. A convenient flag or schema is a candidate representation only if it preserves every typed distinction and source binding here. Deferred representation detail, profile-local policy, an unbound profile, missing source digest, or unavailable evidence is not a pass.

Stop rule. If policy owner, authoritative source, profile version, domain, restriction, exception permission, compatibility, or required producer fact cannot be established, engineering must preserve the resulting unavailable state. It may not infer a policy from a name or current behavior.

Contents

1 Identity, registry, and logical domains

1.1 The object of a decision

An *occurrence* is an exact sample–detector occurrence, not merely a matrix cell. Its identity tuple is

$$\boldsymbol{\iota} = (o, s, t, d_{\mathrm{occ}}, d_{\mathrm{uid}}, a, n, q, k, \pi, \ell),$$

where the components identify observation, scan or coherent segment, sample/time occurrence, detector occurrence, stable detector UID, array, network/group, stream or product role, stage, parent, and lifecycle. The tuple may be represented in any lossless form. Dense row, column, detector, map, or file order is not a substitute for stable external identity. Primary detector timestreams have samples on rows and detectors on columns; cross-product detector association uses the declared UID/occurrence relation. Actual timestamps and gaps, rather than nominal cadence, define time support.

Let F_k denote one immutable, versioned producer-fact set about one exact occurrence or detector at lifecycle generation k. Facts are typed and retain producer, scope, stage, parent, asserted state, source binding, and provenance. A producer-local support or Boolean composite is itself an opaque producer fact with declared owner, inputs, truth-domain rule, missing-state rule, scope, use, and version. VAL consumes it; VAL does not reconstruct it from raw causes.

1.2 Knowledge is not Boolean silence

For a fact applicable to its declared domain, use

$$\mathbf{K} = \{\mathbf{T}, \mathbf{F}, \mathbf{U}, \mathbf{C}\}.$$

T is an authoritative positive assertion; **F** is an authoritative explicit negative assertion; **U** is missing or unknown; and **C** is contradictory, ambiguous, or outside the asserted domain. Fact applicability is recorded separately as applicable, inapplicable, or unknown. For a cause family, **F** is admissible only when the owning producer explicitly asserts absence under a declared complete family. Silence is **U**, never **F**. These symbols state scientific meaning, not an enum or bit layout.

A policy predicate uses the same four knowledge outcomes after authority and domain checks. A predicate can be false because authoritative facts violate a declared restriction; an unavailable fact cannot be coerced to false. A structural contradiction prevents the question from being established. A non-gating contradiction remains in the reasons and is unresolved by the disposition algebra in Section 3.

1.3 Use classes and package-qualified profile names

The shared use-class vocabulary is

$$\begin{aligned}\mathcal{U} = \{&\texttt{independent\_exposure}, \texttt{estimator\_fit},\\ &\texttt{operator\_application}, \texttt{output\_retention},\\ &\texttt{map\_upstream\_admission}, \texttt{response\_companion},\\ &\texttt{empirical\_or\_simulation\_population}, \texttt{diagnostic\_display}\}.\end{aligned}$$

Except for the mandatory registered profile `SCI-VAL:independent_exposure@1`, these labels classify questions but do not supply predicates. Operationally distinct questions use immutable, package-qualified names such as `SCI-PTC:basis_fit_admission`, `SCI-PTC:loading_fit_admission`, and `SCI-PTC:coefficient_qc_population`. The former label `analysis_or_gridding_contribution` is not a v0.1 registry key: `SCI-MAP:map_upstream_admission` names only the upstream question and cannot be mistaken for actual numerical contribution. The package namespace records where a profile is registered; scientific ownership remains the explicit owner field of the immutable registry record.

1.4 Owner-bound profile registry

Let $\mathfrak{R}$ be the immutable profile registry. A registry record is

$$\rho = (I_\rho, v_\rho, U_\rho, O_\rho, S_\rho, h_\rho, \mathcal{D}_\rho, R_\rho, X_\rho, Y_\rho, J_\rho, M_\rho),$$

where I_ρ, v_ρ are registry identity and version; U_ρ is the named use; O_ρ is the actual scientific owner; S_ρ and h_ρ are authoritative source identity and exact version or digest; $\mathcal{D}_\rho$ is the applicability domain and object type; R_ρ is the restriction set; X_ρ states exception permissions and non-exceptionable invariants; Y_ρ assigns every response or uncertainty availability fact its owner-supplied role in the closed set

$$\mathcal{Y} = \{\texttt{structural\_gate}, \texttt{required\_permission}, \texttt{decisive\_exclusion}, \texttt{advisory}\};$$

J_ρ is the compatibility/supersession rule; and M_ρ is missing/conflict behavior. Every scientific element is supplied or approved by O_ρ. The registry records and resolves the binding. It does not make VAL the policy author.

The policy supplied for one evaluation is

$$\mathsf{P}_\rho = (\rho, A_\rho, G_\rho, R_\rho, X_\rho, Y_\rho, M_\rho, \Gamma_\rho, L_\rho).$$

A_ρ is applicability; G_ρ is the structural gate; Γ_ρ is any aggregation specification; and L_ρ binds requested, effective, observation-resolved, learned/resolved, and realized lineage. A reserved name without this complete binding is not a policy and cannot be evaluated as eligible.

When the proposition is an aggregate, its profile ρ_Γ is a separate registry record, not the atomic profile reused at a larger object scope. It binds its own owner, version, domain, operator, threshold, and propagation authority together with the exact compatible atomic source profile identity and version.

1.5 Four independent decision axes

The domains are

$$\begin{aligned}
\mathcal{R} &= \{\text{requested}, \text{not_requested}\},\\
\mathcal{A} &= \{\text{applicable}, \text{inapplicable}, \text{applicability_unknown}\},\\
\mathcal{E} &= \{\text{eligible}, \text{ineligible}, \text{decision_unavailable}\},\\
\mathcal{Z} &= \{\text{realized}, \text{incomplete}, \text{failed}, \text{not_produced}\}.
\end{aligned}$$

A record carries (R, A, E, Z) without aliasing axes. Eligibility is a partial proposition: if no eligibility question is evaluated because no request was made or the profile is known inapplicable, the E slot is uninstantiated, written $\varnothing_E$. This is not a fourth disposition and not `decision_unavailable`. A failed or unproduced artifact likewise does not manufacture a disposition.

1.6 Cause and influence records

Let $\mathcal{G}_k$ be the finite set or directed graph of typed direct causes and inherited influences supplied by producers for F_k. Each node retains producer, family, scope, stage, assertion state, and cause identity. Each influence edge additionally retains source, target, support identity, producer, and

$$(\text{coverage}, \text{epistemic status}) \in \{\text{exact}, \text{conservative}\} \times \{\text{confirmed}, \text{possible}\}.$$

A conservative record includes its over-approximation rule and declared false-positive consequence. It may overstate possible influence but may not be relabeled exact. A use-owner policy determines the consequence for its use; the cause graph carries no universal action.

2 Core, registry, and scientific ownership

2.1 Two layers, one authority boundary

SCI-VAL Core defines the fact and policy envelopes, four-axis logic, knowledge states, deterministic evaluation, cause preservation, immutable lifecycle, aggregation mechanics, and replay semantics. The SCI-VAL Profile Registry binds a complete immutable profile to its real scientific owner and source. It also rejects incomplete, conflicting, or incompatible bindings. Neither layer originates a producer fact or a scientific-use predicate.

The distinction between registry mechanism and policy authority is normative. Registration makes an approved policy discoverable and replayable; it does not transfer ownership, authorize a threshold, or turn a reserved name into a policy. A registry record whose scientific owner or source cannot be established is unavailable, not VAL-owned.

2.2 Why facts, causes, support, and eligibility differ

Finiteness, acquisition validity, an unasserted cause, positive weight, membership in an operator mask, producer-local support, retention, numerical computability, and eligibility for a named use answer different questions. A cause explains a producer-owned condition. A local support is the producer's composite decision for one producer-local operation. Eligibility is a disposition produced by evaluating one profile owned by the named-use authority. A consumer's final numerical admission and product validity remain separate again. No bare flag or universal Boolean can carry this contract.

Authority	Owned scientific content
Producer	Atomic facts and causes; explicit negative assertions; origin, validity, response and uncertainty availability; transformation state; producer-local Boolean composites and supports; direct and transitive influence; parentage and lifecycle.
Named-use owner	Applicability, required facts, decisive predicates, restriction composition, cause-preserving exceptions, aggregation and propagation, thresholds, failure scope, policy version, and compatibility for its exact scientific use.
Profile Registry	Immutable binding of profile identity to the actual owner, authoritative source/version or digest, domain, restrictions, exception permissions, compatibility, and missing behavior. It supplies no predicate.
VAL Core	Shared types; authority and registry checks; knowledge-state and four-axis logic; immutable provenance; order-independent cause preservation; deterministic execution of the exact supplied profile; evaluated record.
Consumer	Estimator-specific numerical support and action; response, covariance, normalization, coaddition, recurrence, or final product validity that it owns.

VAL never originates a PTC fit/application/output/coefficient policy, a MAP admission rule, a producer composite, a threshold, or a global cause ranking. The named-use owner can reuse VAL mechanics and register a profile without transferring scientific authority.

2.3 Producer fact and profile envelopes

A fact envelope is scientifically complete for interchange only when it binds, as applicable:

1. exact identity, product role, stage, parent, and lifecycle;
2. owner and assertion authority, including complete-family authority for an explicit negative cause assertion;
3. origin and required causal source links;
4. distinct structural, detector-binding, numerical/domain, coordinate/time, response, uncertainty, and completion states;
5. direct causes, local support/composite identity, operator-control and edge/context state, representative occurrence, and transitive influence with exact/conservative status; and
6. immutable source, plan, version, and provenance identity.

A source-protection mask, edge guard, fit mask, or other control remains an operator input. It does not become acquisition validity or confidence.

A usable profile envelope must contain every field of ρ in Section 1. The registry verifies exact identity, digest, owner, source, domain, restriction and exception declarations, compatibility, response/uncertainty roles, and supersession; it does not fill a missing field. An aggregate profile additionally binds its distinct aggregate proposition, exact atomic source-profile identity/version, operator, threshold, and propagation authority. Requested, effective, observation-resolved, learned/resolved, and realized profile states remain distinct and flow only forward.

2.4 Mandatory canonical independent-exposure profile

The first mandatory canonical record is `SCI-VAL:independent_exposure@1`. Its scientific owner is the exact owner recorded in the immutable registry binding, not the VAL evaluator. The package namespace identifies the registry, not policy ownership. Its domain is an exact sample–detector occurrence with authoritative representative-source identity and origin.

Within that domain, an exact representative source that is synthesized or replaced is not an original independent astronomical exposure. This restriction is non-exceptionable. A supplied profile that attempts to supersede it is structurally invalid and yields a realized `decision_unavailable` record with reason `policy_invalid` when a record can be produced. It is never evaluated as eligible. A scientifically different proposition must receive a different named use rather than weaken this profile.

Other restrictions for independent exposure remain owner-supplied and must enter through a compatible immutable successor record. Direct replacement does not automatically decide operator application, output retention, diagnostic display, or another exact use.

2.5 Adjacent source-binding register

The exact companion register is `SOURCE_BINDING_REGISTER.md`. The table below is the r0.3 registry snapshot; the companion register is the continuing source-binding authority.

Adjacent-source r0.3 registry snapshot.

Authority	Bound source	Meaning admitted	Change consequence
SCI-RTC	v0.1/r0.9 frozen authority; owner-approved sanitized RTC boundary	Representative source, origin, causes, support, influence precision, response/uncertainty availability, lifecycle	Changed origin semantics requires a new binding and independent-exposure compatibility review.
SCI-CAL	v0.1/r0.3 architecture-frozen rationale; scientific authority not frozen; owner-approved sanitized CAL boundary	Factor/domain, detector join, atmosphere role, response, uncertainty, availability meanings	Conditional input; unresolved change leaves dependent profile unavailable.
SCI-PTC	v0.1/r0.4 frozen authority as recorded by the program index; owner-approved sanitized PTC boundary	Distinct basis/loading fit, application, output, coefficient/QC, response, population, and stage roles	Package-qualified policies remain unbound until exact owner records are registered; revision creates a new binding.
SCI-MAP	v0.1/r0.3 house rationale; scientific authority not frozen; accepted F010/ADR boundary	Upstream admission remains distinct from projection, contribution, supports, response/covariance, coaddition, and final validity	Boundary is admitted; exact MAP profile is unavailable and is not invented by VAL.
ALIGN/AST/TEL	Owner-approved sanitized boundary; no standalone frozen package/version supplied	Time, coordinate, association, origin, synthesis/recomputation, binding, frame, local validity roles	Meaning-only dependency until a versioned owning source is supplied.

Source identity/version or digest is part of profile and replay identity. A new source never silently substitutes for the recorded source.

2.6 Decision record and structural gate

One immutable evaluation record V_k binds:

1. $\boldsymbol{\iota}$, F_k, exact parent and product role;
2. complete ρ, policy lineage, source bindings, and registry state;
3. all four axes, with no eligibility assertion where $E = \varnothing_E$;
4. every required fact and its knowledge/applicability state;
5. every material direct cause and inherited influence;

6. every false, unresolved, conflict, exception, unavailable, and policy-invalid reason;
7. response/uncertainty availability and producer-local supports consumed without reinterpretation;
8. decision stage, object type, applicability domain, action scope, aggregate profile and exact atomic-source-profile binding where applicable, aggregation interchange, and realization/failure reason; and
9. identities for fact set, decision, consumer action, propagated aggregate if any, and successor generation.

The record is always read as "eligible/ineligible under registered profile ρ for use U_ρ, evaluated with SCI-VAL Core." An unqualified *valid* or *eligible* token is scientifically incomplete.

Before evaluating a use predicate, VAL checks identity, stable joins, parentage, fact-set immutability, registry identity/digest, actual policy owner and source, use, profile version, applicability authority, producer authority, scope, compatibility, exceptions, and internal consistency of referenced supports and influence. This gate establishes the scientific question; it does not decide consumer science. Missing, conflicting, ambiguous, out-of-domain, unbound, or policy-invalid material yields applicability unknown and a realized `decision_unavailable` record when possible. The check precedes consumer scientific-state mutation.

After that structural domain is established, a non-gating conflict is not a structural failure. It remains preserved as unresolved evidence: an unrelated known decisive exclusion still establishes `ineligible`; with no such exclusion, a required conflict gives `decision_unavailable`. A conflict about an exception never supplies permission and cannot neutralize the underlying restriction.

2.7 Downstream authority remains downstream

MAP owns geometric incidence, estimator contribution, retained exposure, normalization support, science-policy support, response, covariance, coaddition, companion identity, and final raw map validity. A registered MAP-owned `map_upstream_admission` profile may be evaluated by VAL, but success is necessary and not sufficient for MAP contribution, supported output, or final validity. No exact MAP-owned profile is registered by this v0.1 package. NOI, FLT, BEAM, SRC/MODE, FRUIT, PTC, and other consumers retain their estimator, response, recurrence, and output authority. A finite or locally valid descendant cannot promote an invalid or unavailable parent.

3 Deterministic profile evaluation

3.1 Registry integrity and structural domain

Write $b(\rho, \mathfrak{R})$ for registry binding integrity, $g(\rho, \mathsf{P}_\rho, \mathsf{F}_k)$ for the structural interchange gate, and $a(\rho, \mathsf{P}_\rho, \mathsf{F}_k) \in \mathcal{A}$ for applicability. Binding integrity is true only when registry key/version, actual owner, authoritative source/version or digest, use, domain/object type, restrictions, exception permissions, compatibility, supersession, and missing behavior are complete and mutually consistent. A reserved name is insufficient.

For the canonical independent-exposure profile, let r_{origin} be the contract-level restriction and let $X_{\text{forbidden}}$ contain every attempted exception to it. Then

$$X_{\text{forbidden}} \neq \varnothing \quad \Longrightarrow \quad b(\rho, \mathfrak{R}) = \mathbf{F},$$

with reason `policy_invalid`. This is a structural failure, not an exceptional permission. When representative origin is authoritatively synthesized or replaced and no invalid override is attempted, $r_{\text{origin}} = \mathbf{F}$ and the profile disposition is ineligible.

The general gate succeeds only when $b = \mathbf{T}$ and identity, parent, immutable fact set, producer authority, source bindings, policy lineage, applicability authority, scopes, referenced supports, and influence records are present and consistent. Missing or conflicting structural material makes applicability unknown and the decision unavailable; it is not a use-specific exclusion. Once this structural domain is established, a conflict in an unrelated non-gating predicate is unresolved evidence rather than a structural failure.

3.2 Restriction values and exceptions

For each restriction $r_j \in R_\rho$, the owner-bound profile identifies an applicability predicate, a base permission predicate, and its exception class. After authority checking, contradiction $\mathbf{C}$ in a non-gating predicate is retained and normalized to unresolved **U** for composition. An inapplicable restriction contributes neutral permission **T**. The base permission of an applicable restriction is

$$p_j = \begin{cases} \mathbf{T}, & \text{its permission predicate is authoritatively true,} \\ \mathbf{F}, & \text{its permission predicate is authoritatively false,} \\ \mathbf{U}, & \text{restriction applicability or permission is unknown or conflicting.} \end{cases}$$

Only a resolved, permitted same-profile exception can transform this value:

$$q_j = \begin{cases} \mathbf{T}^X, & \text{that exception authoritatively applies to this exact exceptionable restriction,} \\ p_j, & \text{otherwise, including unknown or conflicting exception applicability.} \end{cases}$$

$\mathbf{T}^X$ composes as **T** but requires the registry/profile identity, exception, underlying restriction, and causes to remain in V_k. An unknown or conflicting exception cannot neutralize an underlying $p_j = \mathbf{F}$; the disposition remains ineligible and the exception conflict is preserved. No exception can be supplied by VAL, another use, or a registry record that prohibits it. A changed exception resolution changes immutable resolved-profile lineage and decision identity.

The normalized permission conjunction is

$\wedge_K$	**T**	**U**	**F**
T	**T**	**U**	**F**
U	**U**	**U**	**F**
F	**F**	**F**	**F**

Known exclusion dominates an unrelated unknown or conflicting non-gating restriction; required unresolved information prevents eligibility when no known exclusion exists. The table is commutative and associative, and every conflict remains in the reason record.

3.3 Disposition function

For a requested record that can be realized, define

$$Q(\rho, \mathsf{P}_\rho, \mathsf{F}_k) = \bigwedge_{r_j \in R_\rho} q_j.$$

The disposition is

$$D(\rho, \mathsf{P}_\rho, \mathsf{F}_k) = \begin{cases} \text{decision\_unavailable}, & b \neq \mathbf{T} \text{ or } g \neq \mathbf{T} \text{ or } a = \text{applicability\_unknown}, \\ \varnothing_E, & b = \mathbf{T},\ g = \mathbf{T},\ a = \text{inapplicable}, \\ \text{ineligible}, & b = \mathbf{T},\ g = \mathbf{T},\ a = \text{applicable},\ Q = \mathbf{F}, \\ \text{decision\_unavailable}, & b = \mathbf{T},\ g = \mathbf{T},\ a = \text{applicable},\ Q = \mathbf{U}, \\ \text{eligible}, & b = \mathbf{T},\ g = \mathbf{T},\ a = \text{applicable},\ Q = \mathbf{T}. \end{cases}$$

All restrictions are evaluated sufficiently to preserve every false, unresolved, conflict, influence, exception, and policy-invalid reason; the precedence does not authorize evidence-erasing short circuit.

Request	Applicability	Profile knowledge	Eligibility	Realization
not requested	not assessed or separately recorded	none evaluated	$\varnothing_E$	not produced
requested	inapplicable	domain known outside profile	$\varnothing_E$	realized
requested	applicability unknown	registry, gate, or applicability unresolved	decision unavailable	realized
requested	applicable	at least one decisive false	ineligible	realized

Request	Applicability	Profile knowledge	Eligibility	Realization
requested	applicable	no false; at least one required unresolved	decision unavailable	realized
requested	applicable	all required true	eligible	realized
requested	any	authoritative artifact not completed	no authoritative eligibility assertion	incomplete, failed, or not produced

A realized decision-unavailable record is a successful scientific account of why no disposition can be made. It is not an execution failure.

3.4 No rescue, cause preservation, and determinism

The cause/influence join $\sqcup$ is graph union on stable, provenance-qualified identities:

$$\mathcal{G}_1 \sqcup \mathcal{G}_2 = \mathcal{G}_2 \sqcup \mathcal{G}_1, \qquad (\mathcal{G}_1 \sqcup \mathcal{G}_2) \sqcup \mathcal{G}_3 = \mathcal{G}_1 \sqcup (\mathcal{G}_2 \sqcup \mathcal{G}_3), \qquad \mathcal{G} \sqcup \mathcal{G} = \mathcal{G}.$$

These equations establish order invariance and idempotence; they do not authorize VAL to combine raw causes into a producer-owned support.

For any applicable restriction multiset S, if $\bigwedge_K S = \mathbf{F}$, then for every additional permission $q \in \{\mathbf{T}, \mathbf{U}, \mathbf{F}\}$,

$$\left(\bigwedge_K S\right) \wedge_K q = \mathbf{F}.$$

Another permission cannot rescue an excluded occurrence. A permitted $\mathbf{T}^X$ is a traceable transformation of the exact restriction by the same owner-bound profile, not an unrelated permission. It cannot transform a non-exceptionable contract invariant.

With exact immutable inputs,

$$\mathsf{V}_k = \mathrm{Eval}_{\mathrm{VAL}}(\mathsf{F}_k, \rho, \mathsf{P}_\rho, \Gamma_\rho).$$

No ambient time, iteration order, hidden default, mutable global state, or consumer side effect may affect V_k. Equal canonical scientific inputs produce equal four-axis states and equal reason/cause graphs. Representation may vary only if it preserves this equality.

For a use owner that explicitly declares and justifies a stricter relation $\rho' \preceq \rho$ on the same domain,

$$\{\mathsf{F} : D(\rho', \mathsf{P}_{\rho'}, \mathsf{F}) = \text{eligible}\} \subseteq \{\mathsf{F} : D(\rho, \mathsf{P}_\rho, \mathsf{F}) = \text{eligible}\}.$$

VAL does not infer strictness or compatibility from names, versions, numbers, or apparent predicate count.

3.5 Exact and conservative causal influence

For exact influence graph $\mathcal{G}$ and producer-declared conservative over-approximation $\widehat{\mathcal{G}}$,

$$\mathcal{G} \subseteq \widehat{\mathcal{G}},$$

under the exact support and approximation-rule identities. Edges in $\mathcal{G}$ are confirmed; additional edges in $\widehat{\mathcal{G}}$ are possible. The profile owner supplies the mapping for a possible edge: permit, exclude, make the disposition unavailable, or request review. Review is action metadata, not a fourth eligibility state; unless the same profile independently resolves admission, the disposition remains `decision_unavailable`. VAL supplies no mapping and never promotes a conservative edge to exact or confirmed.

3.6 Nonretroactive lifecycle

For each exact identity,

$$\mathsf{F}_k \longrightarrow \mathsf{V}_k \longrightarrow \mathsf{A}_k \longrightarrow \mathsf{F}_{k+1} \longrightarrow \mathsf{V}_{k+1},$$

where A_k is the consumer action that used immutable V_k. Later facts or source bindings form a new fact set and decision. A later policy resolution forms distinct resolved-profile lineage and a new decision even when it reuses F_k. Neither path mutates F_k, V_k, or A_k.

Fit-invalid, basis-fit exclusion, loading-fit exclusion, application-available, post-fit output rejection, coefficient/QC population, weight-only noncontribution, and advisory state retain their PTC-owned stage and action scope. Only a PTC-owned fit-support change requires the affected fit or fitted-state invalidation.

3.7 Homogeneous aggregation and anti-circular propagation

An aggregation specification is

$$\Gamma = (\rho_\Gamma, I_{\text{atom}}, v_{\text{atom}}, \Omega, T, \mathbf{n}, N, M, \phi, \tau, b, H, B, \Delta),$$

where ρ_Γ is the aggregate proposition's own complete registered profile; $(I_{\text{atom}}, v_{\text{atom}})$ is its exact atomic source-profile binding; Ω is the exact occurrence population; T is observation/scan/segment/time support; $\mathbf{n}$ contains counts by all four axes; N is the denominator; M is missing-decision treatment; ϕ is the operator; τ is an owner-supplied threshold; b is boundary polarity; H is propagation authority; B is advisory or binding role; and Δ describes learned/data-dependent uncertainty. The registry record ρ_Γ supplies its distinct identity/version, actual scientific owner, authoritative source, aggregate object type and domain, failure behavior, and compatibility. It cannot reuse the atomic profile identity or masquerade as another atomic instance.

For every atomic decision, define the v0.1 compatibility key

$$\chi(\mathsf{V}_i) = (I_{\rho_i}, v_{\rho_i}, k_i, \text{type}(\boldsymbol{\iota}_i), \mathcal{D}_{\rho_i}).$$

Base v0.1 permits aggregation only when

$$\chi(\mathsf{V}_i) = (I_{\text{atom}}, v_{\text{atom}}, k_\Gamma, \text{type}_{\text{atom}}, \mathcal{D}_{\text{atom}}) \qquad \forall i \in \Omega,$$

Thus profile identity/version, lifecycle stage, object type, and applicability domain are exact and homogeneous, and the aggregate registry record binds that exact compatibility key. A heterogeneous multiset is unavailable unless an explicit owner-approved transformation profile maps every input to one common aggregate proposition; base v0.1 defines no such transformation.

For a homogeneous population,

$$\mathsf{V}_{\rho_\Gamma,k} = \text{Eval}_{\text{VAL}}(\text{multiset}\{\mathsf{V}_{i,k} : i \in \Omega\}, \rho_\Gamma, \Gamma).$$

The result retains the aggregate profile identity, exact atomic source-profile binding, every input identity, and aggregation lineage. Permuting the same multiset cannot change the result. Partitioned and unpartitioned aggregation may interchange only when the owner O_{ρ_Γ} declares a sufficient summary σ, associative combine operator $\oplus$, and exact equivalence

$$\sigma(\Omega_1 \uplus \Omega_2) = \sigma(\Omega_1) \oplus \sigma(\Omega_2)$$

for the active missing rule, threshold, polarity, and uncertainty. No equivalence is presumed for arbitrary thresholds or quantiles.

An empty or unknown denominator is inapplicable or unavailable under the supplied owner policy, never a valid zero fraction. Reverse propagation obeys

$$\{\mathsf{V}_{i,k}\}_{i\in\Omega} \longrightarrow \mathsf{V}_{\rho_\Gamma,k} \longrightarrow \mathsf{F}^{\text{prop}}_{i,k+1} \longrightarrow \mathsf{V}_{i,k+1}.$$

The propagated aggregate is a new producer fact at generation $k+1$. It cannot rewrite any $\mathsf{V}_{i,k}$ in its denominator or feed the same generation's $\mathsf{V}_{\rho_\Gamma,k}$. An iterative classification requires a separately authorized bounded lifecycle and stop rule.

3.8 Response, uncertainty, and payload ordering

Response and uncertainty availability are typed facts, distinct from their numerical values and from zero. The named-use owner assigns each fact exactly one role from the closed set below; VAL does not choose or reinterpret it.

Owner-supplied role	Deterministic evaluation consequence
`structural_gate`	Without authoritative satisfaction, the proposition is not established: applicability is unknown and the decision is unavailable.
`required_permission`	Authoritative availability contributes permission; authoritative unavailability excludes; unknown or conflict is unresolved.
`decisive_exclusion`	Authoritative failure or unavailability excludes; authoritative nonfailure is neutral permission; unknown or conflict is unresolved.
`advisory`	The exact state is preserved but contributes neutral permission and cannot determine eligibility.

The restriction conjunction then maps exclusion to `ineligible` and a required unresolved state with no exclusion to `decision_unavailable`. VAL never substitutes identity response or zero uncertainty.

Payload processing is ordered:

1. establish identity, authority, parentage, applicability, and validity without numerically consuming an already invalid payload;
2. exclude a producer- or policy-declared invalid occurrence before arithmetic; and
3. only for an admitted required payload, evaluate its numeric domain. A non-finite value then causes failure or scientific unavailability at the owner-declared scope.

Finiteness never supplies missing admission, response, uncertainty, parent validity, or registry authority.

4 Assumptions, authority limits, and availability

4.1 Declared scientific assumptions

This v0.1 contract assumes that producer authority, profile-owner authority, stable identities, and immutable source bindings can be supplied explicitly. It does not assume that every fact or policy is available. Unavailability is a modeled outcome. It further assumes:

1. producer fact sets, registry records, and evaluated records can be immutable and assigned stable identities;
2. every producer assertion, explicit negative, local support, response, uncertainty status, and influence statement can identify owner, scope, stage, source, and version;
3. a use owner can completely declare applicability, predicate roles, missing behavior, permitted exceptions, failure scope, compatibility, and aggregation without VAL filling scientific gaps;
4. exact timestamps and occurrence populations can be retained where time support matters; and
5. consumer action can bind a realized decision identity rather than an ambient mutable flag.

These are interchange assumptions, not claims about a present implementation.

4.2 Contract-level invariants

The following are boundaries, not VAL-authored policy defaults:

- An exact representative occurrence synthesized by ALIGN or replaced by RTC is not an original independent detector exposure. This invariant is non-exceptionable within `SCI-VAL:independent_exposure@1`.
- Nonrepresentative influence stays visible but has no universal downstream disposition.
- Invalid-payload exclusion precedes numerical consumption; finiteness alone establishes no admission.
- Producer facts and composites retain producer authority; registered policy retains named-use-owner authority; consumer estimators and products retain consumer authority.
- Causes accumulate without erasure. Applicable exceptionable restrictions for one use compose conjunctively in permission unless the same profile contains a permitted, cause-preserving exception.

- A later fact, profile resolution, source binding, aggregate propagation, or consumer stage cannot promote or rewrite its immutable parent decision.

4.3 Current source and profile availability

The exact registered canonical profile is `SCI-VAL:independent_exposure@1`. The package-qualified PTC and MAP names listed in `PROFILE_REGISTRY.md` are reserved but unbound in this package. In particular, no exact MAP-owned `SCI-MAP:map_upstream_admission` profile is available. The MAP boundary is known; its predicates, thresholds, and exceptions are not invented. CAL inputs remain conditional on the source-binding register.

This status is a scientific availability statement, not an implementation finding. A requested evaluation under an unbound profile is `decision_unavailable`; it is not a reason to use a broad name, current software behavior, or a neighboring profile as a substitute.

No aggregate profile is registered by this package. Each future aggregate must bind its own profile identity and owner, exact atomic source profile, operator, threshold, domain, and propagation authority. The base rule permits only homogeneous atomic inputs and supplies no heterogeneous transformation.

4.4 Non-goals and deferred authority

SCI-VAL v0.1 does not select a bit mask, flag layout, enum, table, schema, file format, class, database, serialization, numerical threshold, quality estimator, replacement method, filter, mapmaker, fit objective, uncertainty estimator, producer Boolean expression, or production policy. It does not replace, calibrate, fit, grid, coadd, filter, estimate noise, fit a source or beam, or run recurrence. It does not audit, repair, test, validate, optimize, authorize, or describe an implementation.

It owns neither physical event timing nor missing producer facts; neither PTC basis/loading fit, application, output, coefficient/QC, response, or population policies; neither MAP admission, contribution, normalization, science-policy support, response/covariance, coaddition, or final raw validity; and neither another consumer's estimator admission nor output validity. It defines no global cause precedence, heterogeneous aggregation transformation, automatic detector-to-occurrence propagation, or iterative classification loop.

Enabled polarimetry and numerical auxiliary measured-R scientific execution remain outside active v0.1 authority. This contract makes no assertion of implementation conformity, representation fidelity, observational validation, scientific freeze, production readiness, or adjacent-package readiness.

5 Normative requirements and falsifiable predictions

5.1 Normative requirements

SCI-VAL-REQ-001. For one exact identity, VAL shall accept only an immutable producer-fact set and an exact owner-bound registered profile as scientific inputs and shall return a cause-preserving evaluated record; it shall not originate either side.

SCI-VAL-REQ-002. Producers shall remain authoritative for atomic facts, direct causes, explicit negatives, and producer-local Boolean composites/supports; the named-use owner shall remain authoritative for all scientific predicates and permitted exceptions; the registry shall bind that authority; VAL Core shall own only shared interchange, knowledge-state, provenance, cause-preservation, and deterministic execution semantics.

SCI-VAL-REQ-003. Every decision shall bind stable observation, segment, occurrence, detector occurrence/UID, product role, stage, parent, and lifecycle identities as applicable; dense row, column, or file order shall not substitute for stable identity, and detector joins shall use the declared UID/occurrence relation.

SCI-VAL-REQ-004. Fact interchange shall distinguish authoritative true, authoritative explicit false, unknown/missing, contradictory/out-of-domain, and applicability; missing, unavailable, inapplicable, invalid, non-finite, rejected, disabled, automatic, and numeric zero shall remain distinct.

SCI-VAL-REQ-005. Silence about a cause shall remain unknown. An explicit negative cause assertion shall be accepted only from the owning producer under a declared complete cause family.

SCI-VAL-REQ-006. A producer-local composite/support shall carry its owner, exact inputs, truth-domain rule, missing-state behavior, scope, use, and version and shall be consumed as an opaque authoritative producer fact; VAL shall not reconstruct or reinterpret it from raw causes.

SCI-VAL-REQ-007. All material direct causes and inherited influence shall be accumulated by a commutative, associative, idempotent set/graph join without erasure, ranking, or conversion into a universal action.

SCI-VAL-REQ-008. Every realized decision shall bind exact use, registry key/version, actual policy owner, authoritative source/version or digest, immutable fact-set parent, requested/effective/observation-resolved/ learned/realized lineage, and all reason and lifecycle identities needed for exact replay.

SCI-VAL-REQ-009. The eight shared use-class names shall provide vocabulary only except where an explicit canonical registry record binds an owner-approved invariant. Operationally distinct package questions shall use package-qualified profile names, and no name alone shall supply a predicate, threshold, support, estimator, response, covariance, or final-validity rule.

SCI-VAL-REQ-010. VAL shall complete registry integrity and the mandatory structural interchange gate before any use predicate or consumer scientific-state mutation; unresolved registry binding, identity, parent, policy owner/source/version, compatibility, applicability authority, fact authority, or required scope shall make the decision domain unavailable.

SCI-VAL-REQ-011. Request, applicability, eligibility, and realization shall be independent typed axes. The tokens not requested, inapplicable, decision unavailable, ineligible, and not produced shall never be aliases or silently converted across axes.

SCI-VAL-REQ-012. A not-requested evaluation or a known inapplicable profile shall assert no eligibility proposition. Not requested shall not manufacture a decision, and inapplicable shall not be relabeled decision unavailable.

SCI-VAL-REQ-013. For a requested decision, unknown applicability or an unresolved registry/structural gate shall yield a realized `decision_unavailable` record when a record can be produced; an incomplete, failed, or unproduced artifact shall not itself be called an eligibility disposition.

SCI-VAL-REQ-014. After domain establishment, any known decisive false restriction shall yield ineligible even when an unrelated non-gating required predicate is unknown or conflicting; if none is false and at least one required predicate is unknown or conflicting the result shall be decision unavailable; only all required true restrictions shall yield eligible.

SCI-VAL-REQ-015. Evaluation shall preserve every material false, unknown, contradictory, out-of-domain, influence, exception, source-binding, compatibility, and policy-invalid reason even when one known false already determines the disposition.

SCI-VAL-REQ-016. For one exact profile, all applicable restrictions shall compose conjunctively in permission, equivalently disjunctively in exclusion; an additional permission shall not rescue an excluded occurrence.

SCI-VAL-REQ-017. An override, supersession, or exception shall be effective only when permitted and explicit in the same immutable owner-bound profile. Its identity, resolution, underlying restriction, and causes shall remain in the decision; it shall not apply to a registry-declared non-exceptionable invariant; and a changed exception shall produce distinct resolved-profile lineage and decision identity. Unknown or conflicting exception applicability shall not neutralize the underlying restriction.

SCI-VAL-REQ-018. A permission or eligibility disposition shall not be stored or propagated as a cause.

SCI-VAL-REQ-019. Under the mandatory registered profile `SCI-VAL:independent_exposure@1`, an occurrence whose exact representative source is authoritatively synthesized or replaced shall be ineligible regardless of a finite retained payload or permission under another use.

SCI-VAL-REQ-020. Direct replacement and nonrepresentative influence shall not automatically exclude operator application, output retention, diagnostic display, or another named use; each consequence shall require that use owner's exact registered profile.

SCI-VAL-REQ-021. Influence records shall preserve producer, source, target, support, approximation-rule identity, exact versus conservative coverage, and confirmed versus possible status; conservative influence shall never be silently promoted to exact or confirmed.

SCI-VAL-REQ-022. For possible influence, the owner-bound profile shall explicitly select permit, exclude, scientific unavailability, or review. Review shall remain action metadata and shall not introduce a fourth eligibility disposition.

SCI-VAL-REQ-023. A producer- or policy-declared invalid payload shall be excluded before numerical evaluation. Finiteness, positive weight, mask membership, computability, or retention shall never establish eligibility.

SCI-VAL-REQ-024. If an admitted required payload is non-finite, the affected operation shall fail or become scientifically unavailable at the owner-declared scope; its value shall not be coerced to zero or used to retroactively change admission.

SCI-VAL-REQ-025. Response and uncertainty availability shall be typed separately from numerical response and uncertainty. Unknown or unavailable values shall not be treated as zero or identity. The named-use owner shall assign each fact exactly one role from `structural_gate`, `required_permission`, `decisive_exclusion`, or `advisory` and shall declare failure scope. VAL shall apply the role's deterministic structural, permission, exclusion, or neutral-advisory consequence without inventing another role.

SCI-VAL-REQ-026. PTC-owned basis-fit, loading-fit, fit-excluded/application-available, post-fit output-rejection, coefficient/QC-population, weight-only, and advisory states shall retain distinct stage/action scope. Only a PTC-owned change to the affected fit support shall require that fit or fitted-state invalidation.

SCI-VAL-REQ-027. The lifecycle shall be nonretroactive: $\mathsf{F}_k \to \mathsf{V}_k \to \mathsf{A}_k \to \mathsf{F}_{k+1}$. Later facts, influence, source bindings, or propagated aggregates shall produce a new fact-set and decision identity; later profile resolution shall produce distinct lineage and a new decision. Neither path shall mutate the earlier facts, disposition, reasons, or consumer action.

SCI-VAL-REQ-028. For equal immutable scientific inputs, VAL evaluation shall produce equal four-axis states and equal reason/cause graphs, independent of iteration order, ambient time, hidden defaults, mutable global state, or consumer side effects.

SCI-VAL-REQ-029. Replay shall bind canonical scientific content of the fact set, complete registry record, source bindings, applicability, exception resolution, aggregation, and lifecycle; no unversioned current profile, registry entry, or producer state shall substitute for recorded input.

SCI-VAL-REQ-030. Every occurrence-to-detector or detector-to-occurrence aggregation shall bind its own registered aggregate profile identity/version, actual scientific owner, source, object type and domain, and the exact compatible atomic source-profile identity/version. It shall declare exact population, observation/scan/segment/time support, counts by all four axes, denominator, missing treatment, operator, threshold if any, boundary polarity, propagation authority, advisory/binding role, learned/data-dependent uncertainty, and failure scope. It shall not reuse the atomic profile identity or masquerade as another atomic instance.

SCI-VAL-REQ-031. An aggregate shall retain all atomic decision identities, source bindings, exact atomic-profile lineage, and its distinct aggregate-profile lineage. Missing or failed decisions shall be counted and treated only as specified by the aggregation owner, never silently dropped.

SCI-VAL-REQ-032. An empty or unknown aggregation denominator shall be inapplicable or decision unavailable under the supplied policy and shall never be interpreted as a valid zero fraction.

SCI-VAL-REQ-033. No detector-level aggregate shall automatically apply to every occurrence. Reverse propagation shall require exact population, time support, aggregation, and propagation authority and shall create a new producer fact and lifecycle generation.

SCI-VAL-REQ-034. Aggregation over the same exact homogeneous multiset shall be permutation invariant. Partitioned and unpartitioned aggregation shall be interchangeable only when the owner declares a sufficient summary, associative combine rule, and exact equivalence for the active missing, boundary, threshold, and uncertainty rules.

SCI-VAL-REQ-035. A VAL-evaluated MAP-owned `map_upstream_admission` result shall remain necessary but not sufficient for MAP projection, estimator contribution, normalization or science-policy support, response/covariance, coaddition, supported output, or final raw map validity.

SCI-VAL-REQ-036. A consumer shall not relabel decision unavailable as eligible, erase producer causes, infer a clean cause family from silence, or promote an invalid/unavailable parent because a descendant is finite or locally valid.

SCI-VAL-REQ-037. A conflict in required identity, parentage, registry/profile authority, source/compatibility, applicability, or another structural-gate fact shall yield applicability unknown and decision unavailable before consumer mutation. After domain establishment, a known decisive exclusion shall remain ineligible despite an unrelated conflicting non-gating fact; without such an exclusion, a required non-gating conflict shall yield decision unavailable. Every conflict reason shall be preserved.

SCI-VAL-REQ-038. VAL shall not repair, infer, strengthen, recompute, or default a missing producer fact/support, registry binding, or policy element, including by substituting clean cause, identity response, zero uncertainty, finite payload, source alias, or a policy predicate.

SCI-VAL-REQ-039. Monotonicity shall be asserted only for a use-owner-declared stricter-profile relation on an exact common domain. Under that relation, eligibility under the stricter profile shall imply eligibility under the weaker profile; VAL shall not infer strictness from names, versions, or registry order.

SCI-VAL-REQ-040. Requested, effective, observation-resolved, learned/resolved, and realized profile state shall flow one way and remain in decision lineage; a realized state shall not rewrite its requested or effective ancestors.

SCI-VAL-REQ-041. The contract and any future conformance claim against it shall remain representation-independent and shall not select numerical thresholds, estimators, flags/bit layouts, schemas, producer composites, PTC or MAP scientific policy, heterogeneous transformation policy, or another production default.

SCI-VAL-REQ-042. SCI-VAL v0.1 shall not be used to authorize enabled polarimetry or numerical auxiliary measured-R scientific execution, nor to claim implementation conformity, representation fidelity, observational validation, scientific freeze, production readiness, or adjacent-package readiness.

SCI-VAL-REQ-043. SCI-VAL Core and the Profile Registry shall remain separate authority layers: Core owns shared evaluation semantics; the registry binds immutable owner-approved profiles; neither shall author, approve by inference, duplicate, or acquire the scientific policy it carries.

SCI-VAL-REQ-044. A usable registry record shall bind immutable key and version, named use, actual scientific owner, authoritative source and exact version/digest, domain/object type, restrictions, exception permissions, compatibility/supersession, and missing behavior. A reserved or partial record shall be unavailable.

SCI-VAL-REQ-045. The direct representative synthesis/replacement restriction in `SCI-VAL:independent_exposure@1` shall be non-exceptionable. Any supplied profile that attempts to override it shall be rejected as policy invalid and shall never yield eligible.

SCI-VAL-REQ-046. The former `analysis_or_gridding_contribution` label shall not be accepted as a v0.1 registry key or automatic alias. MAP upstream admission shall use `SCI-MAP:map_upstream_admission`; PTC basis fit, loading fit, application, output, coefficient/QC, response, and population questions shall use distinct package-qualified profile identities.

SCI-VAL-REQ-047. Base v0.1 aggregation shall require every atomic decision to share exact profile identity/version, lifecycle stage, object type, and applicability domain, and the distinct registered aggregate profile shall bind that exact atomic source-profile key. A heterogeneous input shall be unavailable unless an explicit owner-approved transformation profile maps all inputs to one common aggregate proposition; SCI-VAL v0.1 supplies no such policy.

SCI-VAL-REQ-048. An aggregate propagated back to occurrences shall create successor-generation facts and decisions. It shall not rewrite an atomic decision in its own denominator, feed the same generation's aggregate, or create an iterative classification loop without a separately authorized bounded lifecycle and stop rule.

SCI-VAL-REQ-049. Every imported RTC, CAL, PTC, MAP, or other adjacent meaning shall bind the exact approved source/version or digest and compatibility consequence. A changed, missing, or non-authoritative source shall require a new binding or an unavailable result; VAL shall not substitute ambient current meaning or invent an unavailable MAP profile.

5.2 Falsifiable predictions

The following predictions are contract tests, not reports of tests already performed.

SCI-VAL-PRED-001. For a finite, retained occurrence whose exact representative source is authoritatively replaced, every conforming `SCI-VAL:independent_exposure@1` evaluation is ineligible and preserves the replacement cause.

SCI-VAL-PRED-002. If an exact noncanonical diagnostic-display or frozen-operator profile is actually registered, applicable, and satisfied, the same occurrence may be eligible for only that use while its independent-exposure decision remains ineligible and unchanged. An illustrative unregistered profile produces no such decision.

SCI-VAL-PRED-003. If no authoritative package-qualified estimator-fit or MAP upstream-admission profile is bound, a requested evaluation cannot become eligible from finiteness, retention, positive weight, or another use's permission; it is decision unavailable.

SCI-VAL-PRED-004. Every permutation of the same producer-qualified cause/influence graph and exact registered profile produces the same disposition and reason graph.

SCI-VAL-PRED-005. Repeating an identical cause or influence edge does not change disposition, unique-cause counts, or replay identity based on canonical scientific content.

SCI-VAL-PRED-006. With an established decision domain, one known decisive false restriction plus any unrelated unknown or conflicting non-gating restriction yields ineligible and preserves every reason.

SCI-VAL-PRED-007. With an established decision domain, no decisive false and at least one required unknown or conflicting restriction yields decision unavailable; changing that fact to authoritative true can yield eligible only if all other requirements are true.

SCI-VAL-PRED-008. A missing or conflicting exact parent, registry digest, policy source/version, compatibility, or applicability authority yields applicability unknown and decision unavailable before consumer mutation, even when the payload is finite.

SCI-VAL-PRED-009. Not requested and known inapplicable evaluations carry no eligibility proposition; neither is observationally identical to a realized decision-unavailable record.

SCI-VAL-PRED-010. A permitted resolved same-profile exception can neutralize only the exceptionable restriction it names; the result retains exception, restriction, and causes, and an unrelated exclusion still prevents eligibility. An unknown or conflicting exception cannot neutralize its underlying exclusion.

SCI-VAL-PRED-011. The same possible conservative influence may be permitted, excluded, marked for review, or made scientifically unavailable by different authoritative profiles without changing or relabeling the producer influence record.

SCI-VAL-PRED-012. Adding F_{k+1} and evaluating V_{k+1} leaves the serialized scientific content and action association of F_k, V_k, and A_k unchanged.

SCI-VAL-PRED-013. An empty or unknown aggregation denominator never produces an eligible zero-fraction result; the owner-declared outcome is inapplicable or decision unavailable.

SCI-VAL-PRED-014. Permutation of an exact homogeneous aggregate multiset does not alter its result; partitioned evaluation matches unpartitioned evaluation only for an explicitly declared and satisfied sufficient-summary equivalence. The result binds a distinct registered aggregate profile and its exact atomic source-profile identity.

SCI-VAL-PRED-015. A declared-invalid non-finite payload is excluded without numerical consumption; a required payload admitted before discovering non-finiteness instead fails or becomes unavailable at the declared scope. Neither path is converted to eligibility by finiteness logic.

SCI-VAL-PRED-016. Response/uncertainty facts assigned `structural_gate`, `required_permission`, `decisive_exclusion`, and `advisory` produce respectively the declared structural, permission, exclusion, and neutral-preserved outcomes; unknown or unavailable state is never replayed as numeric zero or identity.

SCI-VAL-PRED-017. Upstream eligibility under a MAP-owned profile alone cannot establish MAP numerical contribution, supported output, or final raw map validity.

SCI-VAL-PRED-018. For a correctly declared stricter relation $\rho' \preceq \rho$ on the same domain, no fact set ineligible under ρ can become eligible under ρ'; a counterexample falsifies either the declaration or evaluator.

SCI-VAL-PRED-019. A profile that attempts to admit a directly synthesized or replaced representative occurrence as independent exposure is rejected as policy invalid and produces no eligible result.

SCI-VAL-PRED-020. Two otherwise identical profile records with different authoritative source digests produce distinct replay identities; an absent digest cannot be silently replaced by the current source.

SCI-VAL-PRED-021. A base-v0.1 aggregate containing two profile versions, lifecycle stages, object types, or applicability domains is unavailable before its numerical operator is applied. So is an aggregate whose own registered profile or exact atomic source-profile binding is absent, or whose aggregate identity reuses the atomic identity.

SCI-VAL-PRED-022. Propagating a detector aggregate to its source occurrences creates generation $k+1$ facts and decisions while every generation k denominator decision remains bit-for-bit scientifically unchanged.

SCI-VAL-PRED-023. Supplying only the reserved name `SCI-PTC:basis_fit_admission` or `SCI-MAP:map_upstream_admission` cannot yield eligibility without the actual owner's complete immutable registry record.

SCI-VAL-PRED-024. If an adjacent source revision changes an imported meaning without an approved compatibility binding, the dependent evaluation becomes unavailable rather than replaying under silently updated semantics.

6 Worked cases and edge behavior

6.1 One finite, retained, replaced occurrence

Consider occurrence ω_r with complete stable identity and parentage. Its payload is finite and retained. RTC authoritatively states that the exact representative source was replaced and supplies cause c_{repl} and its source link. Only the canonical independent-exposure profile is registered by this package. Other rows below are evaluable only when the indicated owner binds a complete immutable profile.

Scientific question	Registry/policy state	Contract result
Count as an independent astronomical exposure?	`SCI-VAL:independent_exposure@1` is bound; the direct-origin fact is authoritative	Ineligible. Replacement is non-exceptionable and c_{repl} remains a cause.
Stabilize continuity or receive a frozen operator?	Hypothetical complete owner-bound `operator_application` profile used only for illustration	Would be eligible for that exact application only if that exact profile were registered, applicable, and satisfied; this permission could not rescue independent exposure.
Remain in an output?	Hypothetical complete owner-bound `output_retention` profile used only for illustration	Would be eligible for retention only if that exact profile were registered, applicable, and satisfied; retention would say nothing about fit or map admission.
Be displayed for diagnosis?	Hypothetical complete package-qualified diagnostic profile used only for illustration	Would be eligible for display only if that exact profile were registered, applicable, and satisfied, with replacement provenance visible and no stronger claim.
Enter the PTC basis fit?	Only reserved name `SCI-PTC:basis_fit_admission` is present	Decision unavailable. A name, finiteness, retention, and operator support cannot supply PTC policy.
Enter the PTC loading fit or coefficient/QC population?	No exact owner-bound profile is registered	Decision unavailable separately for each question; one broad estimator-fit answer cannot substitute.
Pass MAP upstream admission?	The accepted MAP boundary is bound, but no exact `SCI-MAP:map_upstream_admission` profile is registered	Decision unavailable. Even later eligibility cannot establish projection, contribution, support, or final map validity.
Receive a response companion?	A hypothetical complete owner profile assigns response availability the `structural_gate` role, but the fact is unknown	Would be decision unavailable if that exact profile were registered and applicable; unknown response is not identity response.

One occurrence is therefore finite, retained, directly replaced, ineligible for independent exposure, conditionally eligible in hypothetical examples only if their exact profiles are later registered and applicable, and undecidable for unbound uses. One Boolean cannot represent these propositions without losing owner, profile, or reason.

6.2 Forbidden independent-exposure exception

Suppose a supplied independent-exposure profile contains an exception saying that a finite replacement may count as an independent exposure. The registry does not first evaluate replacement and then apply the exception. It rejects the profile binding because the attempted exception conflicts with a non-exceptionable contract invariant. The requested result is `decision_unavailable` with `policy_invalid`; it is never eligible. A policy for a genuinely different scientific proposition requires a different named use.

Generic permitted exceptions still behave cause-preservingly. If two exceptionable restrictions exclude an occurrence and an authorized exception supersedes one, the other exclusion remains. If it supersedes the only exceptionable exclusion and all other predicates are true, the result can be eligible while retaining cause, restriction, and exception. If exception applicability is unknown or conflicting, no supersession has been established: the underlying exclusion remains decisive and the conflict is retained.

6.3 Restriction and missing-fact cases

Knowledge state	Disposition	Preserved evidence
all required true	eligible	all inputs, source bindings, profile identity
one false, all others true	ineligible	false restriction and causes
one false, another unknown	ineligible	both false and unknown reasons
one false, unrelated non-gating conflict	ineligible	false and conflict reasons
none false, one unknown	decision unavailable	every unresolved reason
none false, required non-gating conflict	decision unavailable	every conflict reason
structural registry/source conflict	applicability unknown; decision unavailable	conflict; no consumer mutation
false restriction, conflicting exception applicability	ineligible	restriction, exception conflict, and causes
known inapplicable profile	no eligibility proposition	applicability and lineage
not requested	no eligibility proposition	request state; no decision invented
artifact production fails	no authoritative eligibility assertion	realization failure, not disposition

6.4 Silence, explicit absence, and conflicting producers

Suppose a profile requires authoritative absence of cause family K. No record gives **U** and therefore decision unavailable when no other restriction is false. Explicit absence from the owning producer is **F** for cause presence only when the producer declares K complete; that can make an "absence established" predicate true. An assertion from a non-owner or a conflict between authoritative records does not establish cleanliness. The conflict is preserved and the decision fails closed.

6.5 Direct origin versus transitive influence

Direct representative replacement triggers the independent-exposure invariant. A nonrepresentative occurrence connected by an exact confirmed influence edge is governed by the registered profile for its use. A conservative possible edge is different again: its approximation rule and false-positive consequence remain visible, and the profile may permit, exclude, request review, or return decision unavailable. No choice changes the producer-owned edge or imposes the same choice on another use.

6.6 PTC stage separation and package-qualified questions

PTC-owned question/state	Immediate scope	Forbidden inference
basis-fit admission	basis learning population	does not define loading-fit or application support
loading-fit admission	loading-model population	does not rewrite basis-fit membership
fit excluded, application available	fit versus resolved operator application	fit exclusion alone does not forbid application
post-fit output reject	output after fit	does not rewrite earlier fit
coefficient/QC population	coefficient or QC estimation	is not an alias for either fit population
weight-only noncontribution	declared numerical role	zero contribution is not universal invalidity
advisory only	review/provenance	advice is not eligibility

Only the PTC owner can bind these profiles. A later support change forms F_{k+1} and V_{k+1}; only the affected PTC fit-support change requires refit or fitted-state invalidation.

6.7 Aggregation compatibility and feedback

An aggregate containing decisions from `diagnostic_display` and `independent_exposure`, from two profile versions, from different lifecycle stages, or from sample and detector object types is not a meaningful base-v0.1 fraction. The compatibility gate returns unavailable before applying the numerical operator. A future transformation could define one common proposition, but only its scientific owner can authorize it.

For a homogeneous population, unknown decisions cannot disappear from the denominator unless the aggregation owner explicitly states their treatment. The aggregate also has its own registered profile identity, owner, version, domain, operator, threshold, propagation authority, and exact atomic source-profile binding. It cannot reuse the atomic identity or masquerade as another occurrence evaluated under the atomic profile. A missing aggregate profile makes the aggregate proposition unavailable before arithmetic. Empty support is not a scientifically good detector with zero rejected fraction. If a detector aggregate is propagated back to its occurrences, the propagated result becomes a new generation of producer facts. The original atomic decisions and denominator remain immutable. A same-generation loop is rejected; an iterative classifier needs a separately authorized bounded lifecycle and stop rule.

6.8 Response and payload edges

Response and uncertainty facts can be available, known unavailable, unknown, or inapplicable without becoming numeric zero. The owner-bound profile assigns exactly one role: `structural_gate` makes unresolved or unsatisfied structure applicability unknown and the decision unavailable; `required_permission` requires authoritative permission; `decisive_exclusion` makes authoritative failure ineligible; and `advisory` preserves the state without deciding eligibility. Unknown or conflicting required permission/exclusion is unresolved. The role and scope come from the owner-bound profile, not VAL.

An invalid payload—finite or non-finite—is excluded before arithmetic. If an admitted required payload is then found non-finite, the operation fails or becomes unavailable at its declared scope. These orders are observationally different and remain in provenance.

A Informative conformance-record schema

This appendix supplies bookkeeping guidance only. It adds no scientific requirement. One future evidence record should bind one primary claim layer and the exact clauses it addresses.

Field	Evidence content
Record identity	Stable review ID, date, reviewer role, exact contract revision, and supersession status
Claim layer	One primary layer: logical/evaluator conformity, representation fidelity, observational performance, science impact, or production readiness
Target identity	Exact source/build/artifact/configuration identity needed to reproduce the observation
Authority binding	Requirement or prediction IDs, exact profile registry key/version, scientific owner, authoritative source/version or digest
Preconditions	Object identity, fact-set parent, domain, lifecycle stage, profile binding, aggregation compatibility key, and open decisions
Method	Static trace, deterministic fixture, property test, serialization inspection, failure injection, or preregistered observational method
Expected result	Exact invariant and tolerance justified before observing candidate output
Observed result	Values, typed states, messages, artifacts, and exceptions without interpretation hidden in a pass flag
Disposition	Pass, fail, not exercised, blocked by owner decision, or evidence unavailable
Claim limit	What the record does not establish

No status may be inferred from an omitted record. Passing a fixture supports only the named requirement and claim layer; it does not freeze science or establish production readiness.

B Informative review routing

Clause range	Primary review surface	Useful evidence classes
REQ-001–010	Ownership, identity, fact and registry envelopes	Exact-owner and digest fixtures; incomplete-profile rejection; stable-join and complete-family-negative tests
REQ-011–018	Four axes and disposition algebra	Exhaustive truth-domain, not-requested/inapplicable, false-plus-unknown, no-rescue, and cause-preserving exception fixtures
REQ-019–026	Direct origin, influence, payload, response, PTC stages	Representative replacement; exact/conservative edge; invalid-before-numeric; package-qualified stage-separation fixtures
REQ-027–029	Lifecycle, determinism, replay	Snapshot immutability, permutation/idempotence, source-digest and profile-version replay
REQ-030–034	Aggregation	Homogeneous compatibility key, denominator, missing-decision, permutation, and declared partition-equivalence tests
REQ-035–042	Consumer boundaries and claim limits	MAP admission-versus- contribution separation; no-promotion; fail-closed conflict; no-default tests
REQ-043–049	Core/registry repair	Policy-authority separation, non-exceptionable invariant, naming migration, heterogeneous rejection, anti-circular propagation, and source-change fixtures
PRED-001–024	Contract falsifiers	One explicit counterexample-oriented fixture per applicable prediction

The exact Markdown crosswalk is the one-to-one traceability authority. This routing table merely groups evidence work and cannot replace it.